

EE 5239 Nonlinear Programming

Lecture 13b: Sparse and Low Rank Optimization

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Outline

- 1 Review
- 2 Proximal Gradient Descent
- 3 The Block Coordinate Proximal Gradient
- 4 The Group LASSO problem

Last Time

- Applications of structured optimization/learning
- Mathematical models to induce sparsity
- Most of the structured optimization problem is given by

$$\min f(\mathbf{x}) := \underbrace{L(\mathbf{x})}_{\text{data fidelity term}} + \underbrace{R(\mathbf{x})}_{\text{regularization}}$$

- Alternatively, they can be formulated as

$$\min f(\mathbf{x}) := L(\mathbf{x}), \text{ s.t. } R(\mathbf{x}) \leq s$$

- To induce structure, the regularization is usually some kind of nonsmooth “norm”

Why ℓ_1 norm?

- Consider the following scalar optimization problem

$$\min \frac{1}{2}|x - b|^2 + \lambda|x| \quad (P)$$

where $\lambda > 0$ is some coefficient, $x \in \mathbb{R}$

- First order optimality $0 \in \partial f(x)$ (recall subgradient of $|x|$)

$$\partial f(x) = \begin{cases} (x - b) - \lambda & \text{if } x < 0 \\ (-b) + \lambda\epsilon & \text{if } x = 0 \\ (x - b) + \lambda & \text{if } x > 0 \end{cases} \quad (0.1)$$

where $\epsilon \in [-1, 1]$

- The solution becomes

$$x^* = \begin{cases} b + \lambda & \text{if } b + \lambda < 0 \\ 0 & \text{if } |b| \leq \lambda \\ b - \lambda & \text{if } b - \lambda > 0 \end{cases} \quad (0.2)$$

Why ℓ_1 norm

- The optimal solution for problem (P)
- The solution becomes the so-called **Soft-thresholding** operation

$$x^* = \begin{cases} b + \lambda & \text{if } b + \lambda < 0 \\ 0 & \text{if } |b| \leq \lambda \\ b - \lambda & \text{if } b - \lambda > 0 \end{cases} \quad (0.3)$$

- Define $x^* = \text{Soft}(b, \lambda)$; graphically...

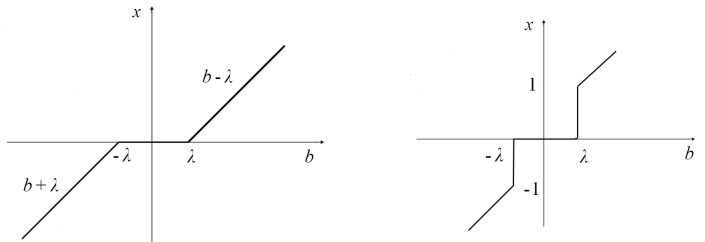


Figure 0.1: Left: The soft thresholding. Right: The hard thresholding.

This Time

- Continue our discussion on structured optimization/learning
- Focusing on algorithms (generalization of GD, BCD, etc)
- Will talk about specific ways to solve popular models such as LASSO, Group LASSO, etc

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How to Solve Structured Sparse Optimization?

- First consider the following problem (unconstrained min)

$$\min f(\mathbf{x}) := g(\mathbf{x}) + R(\mathbf{x})$$

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- **Question:** Can we perform “Gradient Descent” type of update?

Proximal Gradient Descent Method

- Recall in GD we approximate the smooth function g by its second order approximation (upper-bound)

$$g(\mathbf{x}) \leq g(\mathbf{x}^r) + \langle \nabla g(\mathbf{x}^r), \mathbf{x} - \mathbf{x}^r \rangle + \frac{L}{2} \|\mathbf{x} - \mathbf{x}^r\|^2$$

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- Adding both sides by $R(\mathbf{x})$ we have

$$g(\mathbf{x}) + R(\mathbf{x}) \leq \underbrace{g(\mathbf{x}^r)}_{\text{constant}} + \underbrace{\langle \nabla g(\mathbf{x}^r), \mathbf{x} - \mathbf{x}^r \rangle + \frac{L}{2} \|\mathbf{x} - \mathbf{x}^r\|^2 + R(\mathbf{x})}_{\text{simple quadratic function} + \text{Nonsmooth term}}$$

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- Gradient descent finds $\mathbf{x}^{r+1}$ by **minimizing the upper-bound**
- Can we do the same here?**

Proximal Gradient Descent Method

- Minimizing the right-hand side with respect to $\mathbf{x}$ yields

$$\begin{aligned}\mathbf{x}^{r+1} &= \arg \min_{\mathbf{x}} \langle \nabla g(\mathbf{x}^r), \mathbf{x} - \mathbf{x}^r \rangle + \frac{L}{2} \|\mathbf{x} - \mathbf{x}^r\|^2 + \lambda R(\mathbf{x}) \\ &= \arg \min_{\mathbf{x}} \frac{L}{2} \|\mathbf{x} - (\mathbf{x}^r - \frac{1}{L} \nabla g(\mathbf{x}^r))\|^2 + \lambda R(\mathbf{x}) \\ &= \arg \min_{\mathbf{x}} \frac{1}{2} \|\mathbf{x} - (\mathbf{x}^r - \frac{1}{L} \nabla g(\mathbf{x}^r))\|^2 + \frac{\lambda}{L} R(\mathbf{x}) \\ &:= \text{prox}_{\frac{\lambda}{L} R} \left(\mathbf{x}^r - \frac{1}{L} \nabla g(\mathbf{x}^r) \right)\end{aligned}$$

in the second equality we just complete the squares

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in the second equality we just complete the squares

- The solution to the right hand side is called **the proximity operator**

$$\text{prox}_{\mathbf{R}}^{\beta}(\mathbf{y}) := \arg \min_{\mathbf{x}} \frac{1}{2} \|\mathbf{x} - \mathbf{y}\|^2 + \beta R(\mathbf{x})$$

The Properties of Proximity Operator

$$\text{prox}_R^\beta(\mathbf{y}) := \arg \min_x \frac{1}{2} \|\mathbf{x} - \mathbf{y}\|^2 + \beta R(\mathbf{x})$$

- Is this problem easy to solve?
- Think about the case where $R(x) = \gamma \|x\|_1$
- We will come back to this later

The Properties of Proximity Operator

$$\text{prox}_R^\beta(\mathbf{y}) := \arg \min_{\mathbf{x}} \frac{1}{2} \|\mathbf{x} - \mathbf{y}\|^2 + \beta R(\mathbf{x})$$

- If $R(\mathbf{x}) = \iota_X(\mathbf{x})$ (the indicator function of a convex set X), then

$$\text{prox}_{\iota_X}^\beta(\mathbf{y}) = \text{proj}_X(\mathbf{y})$$

where the indicator function is given by

$$\iota_X(\mathbf{x}) = \begin{cases} 1 & \text{if } \mathbf{x} \in X \\ \infty & \text{otherwise} \end{cases}$$

- We can view the indicator function as a “regularizer”!

The Properties of Proximity Operator

$$\text{prox}_R^\beta(\mathbf{y}) := \arg \min_x \frac{1}{2} \|\mathbf{x} - \mathbf{y}\|^2 + \beta R(\mathbf{x})$$

- Projection operator is a special case of proximity operator
- Gradient projection method is a special case of proximal gradient method
- Proximity operators also is non-expansive

$$\|\text{prox}_R^\beta(\mathbf{y}) - \text{prox}_R^\beta(\mathbf{z})\| \leq \|\mathbf{y} - \mathbf{z}\|, \forall \mathbf{z}, \mathbf{y}$$

Proximity and Projection Operators

$$\text{prox}_R^\gamma(\mathbf{y}) = \arg \min_{\mathbf{x}} \frac{1}{2} \|\mathbf{x} - \mathbf{y}\|^2 + \gamma R(\mathbf{x}), \quad R(\mathbf{x}) : \text{regularization}$$

$$\text{proj}_X(\mathbf{y}) = \arg \min_{\mathbf{x}} \frac{1}{2} \|\mathbf{x} - \mathbf{y}\|^2 + \iota_X(\mathbf{x}), \quad \iota_X(\mathbf{x}) : \text{indicator function}$$

Key: Easy computation

The Properties of Proximity Operator

Commonly Used Proximity Functions

$\psi(\mathbf{x})$	$\text{prox}_{\psi}(\mathbf{x})$
$\varphi(\mathbf{x} - \mathbf{z}), \mathbf{z} \in \mathbb{R}^n$	$\mathbf{z} + \text{prox}_{\varphi}(\mathbf{x} - \mathbf{z})$
$\varphi(\mathbf{x}/\rho), \rho \in \mathbb{R} \setminus \{0\}$	$\rho \text{prox}_{\varphi/\rho^2}(\mathbf{x}/\rho)$
$\varphi(-\mathbf{x})$	$-\text{prox}_{\varphi}(-\mathbf{x})$
$\varphi(\mathbf{x}) + \alpha \ \mathbf{x}\ ^2/2 + \beta \mathbf{u}^\top \mathbf{x} + \gamma$ $\mathbf{u} \in \mathbb{R}^n, \alpha > 0, (\beta, \gamma) \in \mathbb{R}^2$	$\text{prox}_{\varphi/(\alpha+1)}((\mathbf{x} - \beta \mathbf{u})/(\alpha + 1))$
$\varphi^*(\mathbf{x})$	$\mathbf{x} - \text{prox}_{\varphi} \mathbf{x}$
$d_X^2(\mathbf{x})/2$	$(\mathbf{x} + \text{proj}_X(\mathbf{x}))/2$
$\gamma \ \mathbf{L}\mathbf{x} - \mathbf{y}\ ^2/2$ $\mathbf{L} \in \mathbb{R}^{m \times n}, \gamma > 0, \mathbf{y} \in \mathbb{R}^m$	$(\mathbf{I} + \gamma \mathbf{L}'\mathbf{L})^{-1}(\mathbf{x} + \gamma \mathbf{L}'\mathbf{y})$
$\iota_X(\mathbf{x}) = \begin{cases} 0 & \text{if } \mathbf{x} \in X \\ +\infty & \text{otherwise} \end{cases}$	$\text{proj}_X(\mathbf{x})$
$\gamma d_X(\mathbf{x}), \gamma > 0$	$\begin{cases} \mathbf{x} + \frac{\gamma}{d_X(\mathbf{x})}(\text{proj}_X(\mathbf{x}) - \mathbf{x}) & \text{if } d_X(\mathbf{x}) > 0 \\ \text{proj}_X(\mathbf{x}) & \text{otherwise} \end{cases}$
$\sigma_X(\mathbf{x}) = \sup_{\mathbf{u} \in X} \mathbf{x}'\mathbf{u}$	$\mathbf{x} - \text{proj}_X(\mathbf{x})$
$\sum_{k=1}^N \phi_k(\mathbf{x}^\top \mathbf{b}_k)$ ϕ_k convex, $(\mathbf{b}_k)_{1 \leq k \leq N}$ basis of $\mathbb{R}^n$	$\sum_{k=1}^N \text{prox}_{\phi_k}(\mathbf{x}^\top \mathbf{b}_k) \mathbf{b}_k$

Figure 0.1: A Useful Table (The constant L is fixed at 1).

Prox Operator: Example 1

- Take $R(\mathbf{x}) = \frac{1}{2}d_X^2(\mathbf{x})$, where $d_X(\mathbf{x})$ denotes the distance between $\mathbf{x}$ and a convex set X
- That is, $d_X(\mathbf{x}) = \min_{\mathbf{z} \in X} \|\mathbf{x} - \mathbf{z}\|$
- Then its prox operator is

$$\begin{aligned}\text{prox}_R^1(\mathbf{y}) &= \arg \min_{\mathbf{x}} \frac{1}{2}d_X^2(\mathbf{x}) + \frac{1}{2}\|\mathbf{x} - \mathbf{y}\|^2 \\ &= \arg \min_{\mathbf{z} \in X, \mathbf{x}} \frac{1}{2}\|\mathbf{z} - \mathbf{x}\|^2 + \frac{1}{2}\|\mathbf{x} - \mathbf{y}\|^2 \\ &\stackrel{(i)}{=} \frac{1}{2} \left(\mathbf{y} + \arg \min_{\mathbf{z} \in X} \frac{1}{4}\|\mathbf{z} - \mathbf{y}\|^2 \right) \\ &= \frac{\mathbf{y} + \text{proj}_X(\mathbf{y})}{2}\end{aligned}$$

where (i) is because $\mathbf{x} = \frac{\mathbf{y} + \mathbf{z}}{2}$

- Geometrically what does this mean?

Prox Operator: Example 2

- Take $R(\mathbf{x}) = \gamma \|\mathbf{x}\|_1$, the ℓ_1 norm

Prox Operator: Example 2

- Take $R(\mathbf{x}) = \gamma \|\mathbf{x}\|_1$, the ℓ_1 norm
- We have the following (Note: Separability!)

$$\begin{aligned}\text{prox}_R^\gamma(\mathbf{y}) &= \arg \min_{\mathbf{x}} \gamma \|\mathbf{x}\|_1 + \frac{1}{2} \|\mathbf{x} - \mathbf{y}\|^2 \\ &= \arg \min_{\mathbf{x}} \sum_{k=1}^K \gamma |x_k| + \frac{1}{2} (x_k - y_k)^2\end{aligned}$$

Prox Operator: Example 2

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- The k th component of $\text{prox}_R^\gamma(\mathbf{y})$ is given by

$$[\text{prox}_R^\gamma(\mathbf{y})]_k = \arg \min_{x_k} \gamma |x_k| + \frac{1}{2} (x_k - y_k)^2 = \text{Soft}(y_k, \gamma)$$

Recall that $\text{Soft}(y_k, \gamma)$ is given by

$$x_k^* = \begin{cases} y_k + \gamma & \text{if } y_k + \gamma < 0 \\ 0 & \text{if } |y_k| \leq \gamma \\ y_k - \gamma & \text{if } y_k - \gamma > 0 \end{cases} \quad (0.1)$$

Linear Regression Revisited: The LASSO

- Now let us revisit our linear regression problem
- Consider the following ℓ_1 regularized linear regression

$$\min \frac{1}{2} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 + \lambda \|\mathbf{x}\|_1$$

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- Use Proximal Gradient Method to solve it?
- Yes, because the proximity operator can be computed in **closed-form** (by performing **component-wise soft-thresholding**)

Linear Regression Revisited: The LASSO

- **Question:** How to pick the coefficient λ ?

Linear Regression Revisited: The LASSO

- **Question:** How to pick the coefficient λ ?
- **Popular method:** K-fold Cross-Validation
 - 1 Divide training data equally into K parts (or folds)
 - 2 Pick $K - 1$ parts, do training on a particular λ , test the performance on the “left out” part
 - 3 Do this K times, and average the performance
 - 4 Pick the λ that has the best “averaged” performance

The Proximal Gradient for LASSO

- Recall for ℓ_1 norm the k th component of $\text{prox}_R^\gamma(\mathbf{y})$ is given by

$$\begin{aligned} [\text{prox}_R^\gamma(\mathbf{y})]_k &= \arg \min_{x_k} \gamma |x_k| + \frac{1}{2}(x_k - y_k)^2 \\ &= \text{Soft}(y_k, \gamma) \end{aligned}$$

The Proximal Gradient for LASSO

- Proximal Gradient Descent (PGD) for LASSO

- 1 Start at an initial solution $\mathbf{x}^0$, let $r = r + 1$
- 2 Perform a gradient step **without** $\|\mathbf{x}\|_1$

$$\mathbf{c}^{r+1} = \mathbf{x}^r - \frac{1}{L} \nabla g(\mathbf{x}^r) = \mathbf{x}^r - \frac{1}{L} \mathbf{A}^T (\mathbf{A} \mathbf{x}^r - \mathbf{b})$$

- 3 Generate the next iterate

$$\mathbf{x}^{r+1} = \text{prox}_{\underbrace{\lambda/L}_{\mathbf{c}^{r+1}}}(\mathbf{x}^r - \frac{1}{L} \nabla g(\mathbf{x}^r)),$$

where the prox operator is computed by

$$x_k^{r+1} = \begin{cases} c_k^{r+1} + \lambda/L & \text{if } c_k^{r+1} + \lambda/L < 0 \\ 0 & \text{if } |c_k^{r+1}| \leq \lambda/L \\ c_k^{r+1} - \lambda/L & \text{if } c_k^{r+1} - \lambda/L > 0 \end{cases} \quad (0.2)$$

The Proximal Gradient for LASSO

- Can be viewed as **regular GD** for least square + an extra step of **componentwise soft-thresholding**
- **Main message**: the computation of PGD for LASSO is simple, closed-form
- Two step algorithm:
 - 1 Compute usual gradient w.r.t. the smooth function;
 - 2 Soft-thresholding on the result (where the sparsity comes from)
- **Intuition**: keep shrinking the gradient steps
- **Exercise**: Try the PGD on a randomly generated regression problem

The Proximal Gradient for Sparse LR

- Similarly, we can use PGD to solve the sparse logistic regression

$$\min \frac{1}{M} \sum_{i=1}^M \log(1 + \exp(-b_i \mathbf{a}_i^T \mathbf{x})) + \lambda \|\mathbf{x}\|_1$$

- PGD for Sparse Logistic Regression

- ① Start at an initial solution $\mathbf{x}^0$, let $r = r + 1$
- ② Perform a gradient step **without** $\|\mathbf{x}\|_1$

$$\mathbf{c}^{r+1} = \mathbf{x}^r - \frac{1}{L} \nabla g(\mathbf{x}^r) = \mathbf{x}^r + \frac{1}{LM} \sum_{i=1}^M \frac{b_i \mathbf{a}_i}{1 + e^{b_i \mathbf{a}_i^T \mathbf{x}}}$$

- ③ The next iterate $\mathbf{x}^{r+1} = \text{prox}^{\lambda/L}_R(\mathbf{x}^r - \frac{1}{L} \nabla g(\mathbf{x}^r))$, is computed by

$$x_k^{r+1} = \begin{cases} c_k^{r+1} + \lambda/L & \text{if } c_k^{r+1} + \lambda/L < 0 \\ 0 & \text{if } |c_k^{r+1}| \leq \lambda/L \\ c_k^{r+1} - \lambda/L & \text{if } c_k^{r+1} - \lambda/L > 0 \end{cases} \quad (0.3)$$

Convergence Rate Estimate

- We are interested in the number of iterations requires to obtain ϵ optimal solution, i.e., what is the smallest r that satisfying

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- Define D as the distance between the first iterates and the global optimal solution set

$$D := \max\{\|\mathbf{x}^0 - \mathbf{x}^*\| \mid \mathbf{x}^* \in \mathbf{X}^*\}$$

Convergence Rate Estimate

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$$D := \max\{\|\mathbf{x}^0 - \mathbf{x}^*\| \mid \mathbf{x}^* \in \mathbf{X}^*\}$$

- Convergence rate:

$$f(\mathbf{x}^r) - f(\mathbf{x}^*) \leq \frac{4LD^2}{r+2}, \forall r \geq 1$$

- So reach ϵ optimal solution requires $r \leq \frac{4LD^2}{\epsilon} - 2$

Illustration



Figure 0.2: Illustration of the rate.

Illustration



Figure 0.3: Illustration of the rate.

Illustration



Figure 0.4: Illustration of the rate.

Illustration

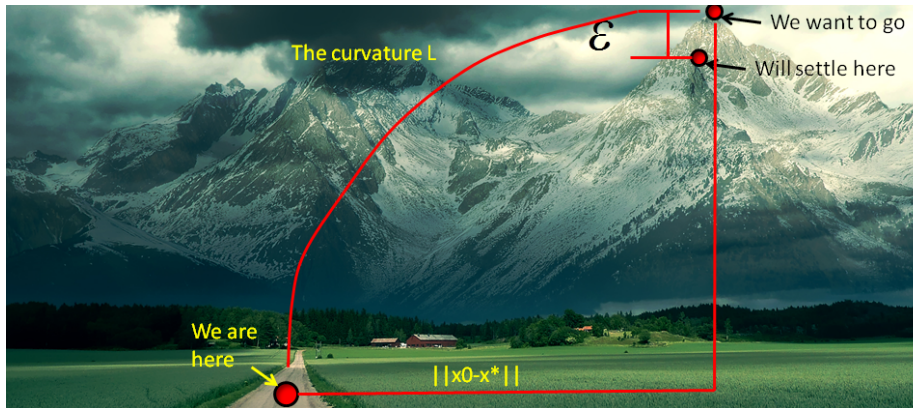


Figure 0.5: Illustration of the rate.

Codes

- To see how LASSO problem can be solved using proximal gradient method, please refer to the link below
- <https://www.cs.ubc.ca/~schmidtm/Software/lasso.html>

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The Block Coordinate Proximal Gradient Method

- Naturally, we can also extend the BCD to nonsmooth case
- Consider the following K -block problem

$$\min_{\{x_k\}} g(x_1, \dots, x_K) + \sum_{k=1}^K \lambda_k R(x_k)$$

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- Consider the following K -block problem

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- Assume that

$$\|\nabla_k g(\mathbf{x}_k, \mathbf{x}_{-k}) - \nabla_k g(\mathbf{y}_k, \mathbf{x}_{-k})\| \leq L_k \|\mathbf{x}_k - \mathbf{y}_k\|, \quad \forall \mathbf{x}_k, \mathbf{y}_k$$

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- A general BCPG algorithm
 - 1 At iteration $r + 1$, pick an index j
 - 2 $\mathbf{x}_j^{r+1} = \text{prox}_R^{\lambda_j/L_j}(\mathbf{x}_j^r - \nabla_j g(\mathbf{x}^r))$
 - 3 $\mathbf{x}_i^{r+1} = \mathbf{x}_i^r, \quad \forall i \neq j$

The Block Coordinate Proximal Gradient Method

- Naturally, we can also extend the BCD to nonsmooth case
- Consider the following K -block problem

$$\min_{\{\mathbf{x}_k\}} g(\mathbf{x}_1, \dots, \mathbf{x}_K) + \sum_{k=1}^K \lambda_k R(\mathbf{x}_k)$$

- Assume that

$$\|\nabla_k g(\mathbf{x}_k, \mathbf{x}_{-k}) - \nabla_k g(\mathbf{y}_k, \mathbf{x}_{-k})\| \leq L_k \|\mathbf{x}_k - \mathbf{y}_k\|, \quad \forall \mathbf{x}_k, \mathbf{y}_k$$

The Block Coordinate Proximal Gradient Method

- A general BCPG algorithm (with exact minimization)
 - ① At iteration $r + 1$, pick an index j
 - ② $\mathbf{x}_j^{r+1} = \arg \min_{\mathbf{x}_j} g(\mathbf{x}_j, \mathbf{x}_{-j}^r) + \lambda_j R(\mathbf{x}_j)$
 - ③ $\mathbf{x}_i^{r+1} = \mathbf{x}_i^r, \forall i \neq j$

The BCPG Method for LASSO

useR! 2009

Trevor Hastie, Stanford Statistics

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Coordinate Descent

- Solve the lasso problem by coordinate descent: optimize each parameter separately, holding all the others fixed. Updates are trivial. Cycle around till coefficients stabilize.
- Do this on a grid of λ values, from λ_{max} down to λ_{min} (uniform on log scale), using warm starts.
- Can do this with a variety of loss functions and additive penalties.

Coordinate descent achieves dramatic speedups over all competitors, by factors of 10, 100 and more.

The BCPG Method for LASSO

- Let us see how to apply BCPG to LASSO
- Let each block consists of a single variable,

$$\min \frac{1}{2} \left\| \sum_{k=1}^K \mathbf{a}_k x_k - \mathbf{b} \right\|^2 + \lambda \sum_{k=1}^K |x_k|_1$$

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- Do the exact minimization version
- Pick block j at iteration $r + 1$

$$\begin{aligned} x_j^{r+1} &= \arg \min \frac{1}{2} \left\| \mathbf{a}_j x_j - \left(- \sum_{k \neq j} \mathbf{a}_k x_k^r + \mathbf{b} \right) \right\|^2 + \lambda |x_j|_1 \\ &= \arg \min \frac{1}{2} (x_j - \langle - \sum_{k \neq j} \mathbf{a}_k x_k^r + \mathbf{b}, \mathbf{a}_j / \|\mathbf{a}_j\|^2 \rangle)^2 + \frac{\lambda}{\|\mathbf{a}_j\|^2} |x_j|_1 \\ &:= \arg \min \frac{1}{2} (x_j - c_j^r)^2 + \lambda_j |x_j|_1 \end{aligned}$$

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- Then is a simple 1-D soft-thresholding operation!

The BCPG Method for LASSO

- **Exercise:** Please derive the PG version of the coordinate descent for LASSO
- What is the iteration complexity for BCPG? Very challenging question, active research area
- Why setting each block to be a single variable?

The BCPG Method for LASSO

- How to find the entire **path** of solutions?
- **Pathwise** Coordinate Descent for LASSO
- Start with large value for λ (very sparse model) and slowly decrease it
- most coordinates that are zero never become non-zero

More on BCPG

- Various different coordinate selection rules
 - ① Cycle through all coordinates (classic)
 - ② Randomly pick the coordinates (recently popular)
 - ③ Pick the coordinates with the **largest gradient component** (may be expensive: parallel implementation?)
 - ④ Pick the one with the **largest objective improvement** (may be expensive; parallel implementation?)
- A useful strategy in practice is **warm start**
- Starting from a big value of λ and gradually reducing the value

Outline

- 1 Review
- 2 Proximal Gradient Descent
- 3 The Block Coordinate Proximal Gradient
- 4 The Group LASSO problem

Group LASSO: Formulations

- Now that we see how to utilize ℓ_1 norm to induce “sparsity” in the desired solutions
- A related problem is how to induce “group-sparsity”; See [HERE](#) for a note by Friedman, Hastie and Tibshirani
- What do I mean by that?!

Group LASSO: Illustration

- We would like to have a group of features to be either active or not active at the same time
- These features are related based on *a priori* empirical information

Diagram illustrating the Group LASSO model. The equation is shown as $y = Ax + z$, where y is an $n \times 1$ vector, A is an $n \times p$ matrix with $p \gg n$, x is a $p \times 1$ vector, and z is an $n \times 1$ vector. The vector x is partitioned into groups, with one group highlighted in a blue box and labeled x^* . Below the diagram, the optimization problem is given as $\min \frac{1}{2} \|Ax - y\|_2^2 + \lambda \sum_{i=1}^g d_i \|x_{G_i}\|_2$. The term $d_i \|x_{G_i}\|_2$ is highlighted in a blue box, with an arrow pointing from the corresponding group in the x vector above.

Group LASSO: Example

- An interesting example is the so-called multi-task learning
- K tasks to be performed at the same time
- But we know that all of them must **pick the same feature**

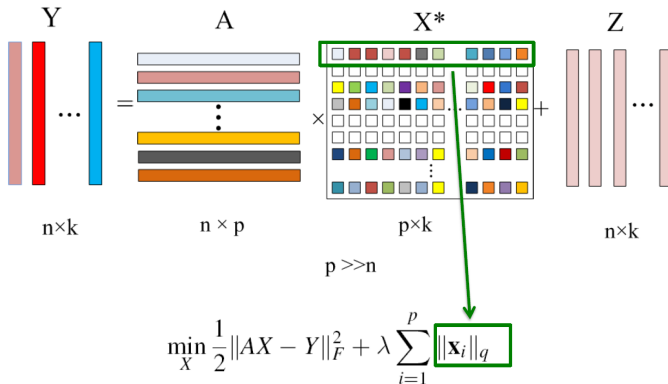


Figure 0.2: Illustration of Multi-Task Learning [J. Ye 10]

Group LASSO: Example

Samples of the letters s and g from the same person



Task	pixels: error (%)			
	ℓ_1/ℓ_2	ℓ_1/ℓ_1	id. ℓ_1	pool
c/e	4.0	8.5	9.0	4.5
g/y	11.4	16.1	17.2	18.6
g/s	4.4	10.0	10.3	6.9
m/n	2.5	6.3	6.9	4.1
a/g	1.3	3.6	4.1	3.6
i/j	12.0	14.0	14.0	11.3
a/o	2.8	4.8	5.2	4.2
f/t	5.0	6.7	6.1	8.2
h/n	3.2	14.3	18.6	5.0

Figure 0.3: Illustration of Multi-Task Learning [Obozinski, Taskar and Jordan 2006]

Group LASSO: Formulation

- Group LASSO has similar formulations as LASSO, just that a **different penalization** is used
- Suppose there are G groups, each has a **subset** of features $\mathbf{x}_g$
- The problem is formulated as

$$\min \frac{1}{2} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 + \sum_{g=1}^G \lambda_g \|\mathbf{x}_g\|_2$$

- The norm $\sum_{g=1}^G \|\mathbf{x}_g\|_2$ is called the ℓ_1/ℓ_2 norm (the ℓ_1 norm of a set of ℓ_2 norms)
- We will assume that the groups are **pre-defined**, and are **separable** (different groups do not overlap)

Group LASSO: Formulation

Diagram illustrating the Group LASSO formulation. The equation is shown as:

$$y = Ax + z$$

where y is $n \times 1$, A is $n \times p$ with $p \gg n$, x is $p \times 1$, and z is $n \times 1$. The vector x is partitioned into groups, with the top group highlighted in a blue box and labeled x^* .

The optimization problem is:

$$\min \frac{1}{2} \|Ax - y\|_2^2 + \lambda \sum_{i=1}^g d_i \|x_{G_i}\|_2$$

The term $d_i \|x_{G_i}\|_2$ is highlighted in a blue box, corresponding to the highlighted group in the vector x .

Figure 0.4: Illustration of Group LASSO [J. Ye 10]

Group LASSO: Why ℓ_1/ℓ_2 norm

- Recall that we have mentioned before that $\|x\|_2$ tends to produce x vector with all zeros

Group LASSO: Why ℓ_1/ℓ_2 norm

- Recall that we have mentioned before that $\|\mathbf{x}\|_2$ tends to produce $\mathbf{x}$ vector with all zeros
- Why? First let us look at the (sub)gradient of

$$\|\mathbf{x}\|_2 := \sqrt{x_1^2 + \cdots, x_K^2}$$

which is

$$\frac{[x_1, \cdots, x_K]^T}{\sqrt{x_1^2 + \cdots, x_K^2}} = \frac{\mathbf{x}}{\|\mathbf{x}\|_2}$$

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- So the gradient exists everywhere except when $\mathbf{x} = \mathbf{0}$
- How about its sub(gradient) at point $\mathbf{x} = \mathbf{0}$?

Group LASSO: Why ℓ_1/ℓ_2 norm

- Use the definition of subgradient $f(\mathbf{x}) - f(\mathbf{y}) \geq \langle \partial f(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle$

Group LASSO: Why ℓ_1/ℓ_2 norm

- Use the definition of subgradient $f(\mathbf{x}) - f(\mathbf{y}) \geq \langle \partial f(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle$
- Let $\mathbf{y} = \mathbf{0}$, and $f(\mathbf{x}) = \|\mathbf{x}\|_2$, we have $\|\mathbf{x}\|_2 \geq \langle \partial \|\mathbf{0}\|_2, \mathbf{x} \rangle$

Group LASSO: Why ℓ_1/ℓ_2 norm

- Use the definition of subgradient $f(\mathbf{x}) - f(\mathbf{y}) \geq \langle \partial f(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle$
- Let $\mathbf{y} = \mathbf{0}$, and $f(\mathbf{x}) = \|\mathbf{x}\|_2$, we have $\|\mathbf{x}\|_2 \geq \langle \partial \|\mathbf{0}\|_2, \mathbf{x} \rangle$
- This means that

$$\partial \|\mathbf{0}\|_2 = \{\mathbf{y} \mid \|\mathbf{y}\|_2 \leq 1\}$$

Group LASSO: Why ℓ_1/ℓ_2 norm

- Then let us look at the proximity operator for $\|\mathbf{x}\|_2$

$$\text{prox}_{\|\mathbf{x}\|_2}^\lambda(\mathbf{z}) = \arg \min \lambda \|\mathbf{x}\|_2 + \frac{1}{2} \|\mathbf{x} - \mathbf{z}\|^2$$

Group LASSO: Why ℓ_1/ℓ_2 norm

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- First order optimality $0 \in -\mathbf{x} + \mathbf{z} - \lambda \partial(\|\mathbf{x}\|_2)$

Group LASSO: Why ℓ_1/ℓ_2 norm

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- If $\|\mathbf{z}\|_2 \leq \lambda$, $\mathbf{x} = \mathbf{0}$

Group LASSO: Why ℓ_1/ℓ_2 norm

- Then let us look at the proximity operator for $\|\mathbf{x}\|_2$

$$\text{prox}_{\|\mathbf{x}\|_2}^\lambda(\mathbf{z}) = \arg \min \lambda \|\mathbf{x}\|_2 + \frac{1}{2} \|\mathbf{x} - \mathbf{z}\|^2$$

- First order optimality $\mathbf{0} \in -\mathbf{x} + \mathbf{z} - \lambda \partial(\|\mathbf{x}\|_2)$
- If $\|\mathbf{z}\|_2 \leq \lambda$, $\mathbf{x} = \mathbf{0}$
- Else, $\mathbf{x}(1 + \frac{\lambda}{\|\mathbf{x}\|_2}) = \mathbf{z}$, or

$$\mathbf{x} = \left(1 + \frac{\lambda}{\|\mathbf{x}\|_2}\right)^{-1} \mathbf{z}$$

Group LASSO: Why ℓ_1/ℓ_2 norm

- If $\|\mathbf{z}\|_2 \leq \lambda$, $\mathbf{x}^* = \mathbf{0}$
- Else, $\mathbf{x}^*(1 + \frac{\lambda}{\|\mathbf{x}^*\|_2}) = \mathbf{z}$, or $\mathbf{x}^* = \left(1 + \frac{\lambda}{\|\mathbf{x}^*\|_2}\right)^{-1} \mathbf{z}$
- Need to know $\|\mathbf{x}^*\|_2$, take norm on both sides

$$\|\mathbf{x}^*\|_2 = \left\| \left(1 + \frac{\lambda}{\|\mathbf{x}^*\|_2}\right)^{-1} \mathbf{z} \right\|_2$$

or we want to find a $\delta > 0$ such that

$$\delta = \left\| \left(1 + \frac{\lambda}{\delta}\right)^{-1} \mathbf{z} \right\|_2$$

- solve δ directly: $\delta = \|\mathbf{z}\| - \lambda$

Group LASSO: PG Method

- Now we can solve group LASSO by proximal gradient method

- 1 Start at a initial solution $\mathbf{x}^0$, let $r = r + 1$
- 2 Perform a gradient step **without regularizers**

$$\mathbf{c}^{r+1} = \mathbf{x}^r - \frac{1}{L} \nabla g(\mathbf{x}^r) = \mathbf{x}^r - \frac{1}{L} \mathbf{A}^T (\mathbf{A} \mathbf{x}^r - \mathbf{b})$$

Decompose $\mathbf{c}^{r+1} = \{\mathbf{c}_g^{r+1}\}$ according to the pre-defined groups

- 3 The next iterate $\mathbf{x}_g^{r+1} = \text{prox}^{\lambda_g/L}_R(\mathbf{x}_g^r - \frac{1}{L} \nabla_g g(\mathbf{x}^r))$:

$$\mathbf{x}_g^{r+1} = \begin{cases} 0 & \text{if } \|\mathbf{c}_g^{r+1}\| \leq \lambda_g/L \\ (1 + \frac{\lambda_g}{L\delta_g}) \mathbf{c}_g^{r+1} & \text{otherwise} \end{cases} \quad (0.1)$$

where δ_g is computed in the previous slide, for each group g .

- The difference between this solution and the LASSO solution?

Group LASSO: BCD Method

- BCD method is also ideal for group lasso
- The objective of group LASSO is given by

$$\frac{1}{2} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 + \sum_{g=1}^G \lambda_g \|\mathbf{x}_g\|_2 = \frac{1}{2} \left\| \sum_g \mathbf{A}_g \mathbf{x}_g - \mathbf{b} \right\|^2 + \sum_{g=1}^G \lambda_g \|\mathbf{x}_g\|_2$$

- The key is to solve the subproblem of the following form

$$\frac{1}{2} \|\mathbf{A}_g \mathbf{x}_g - \mathbf{c}\|^2 + \lambda_g \|\mathbf{x}_g\|_2$$

- Again closed form solution (subject to bisection search)

if $\|\mathbf{A}_g \mathbf{c}\| \leq \lambda_g$, then $\mathbf{x}_g^* = \mathbf{0}$

if $\|\mathbf{A}_g \mathbf{c}\| > \lambda_g$, then $\mathbf{x}_g^* = (\mathbf{A}_g^T \mathbf{A}_g + \delta \lambda_g \mathbf{I})^{-1} \mathbf{A}_g^T \mathbf{c}$

with $\delta > 0$ satisfying $\delta^{-1} = \|(\mathbf{A}_g^T \mathbf{A}_g + \delta \lambda_g \mathbf{I})^{-1} \mathbf{A}_g^T \mathbf{c}\|$

Other Types of Sparse Modeling

- There are many other types of sparse modeling
- All of them utilize *a priori* information about the problem
- Here we briefly mention a few of them without going into too much details

① Sparse Group LASSO:

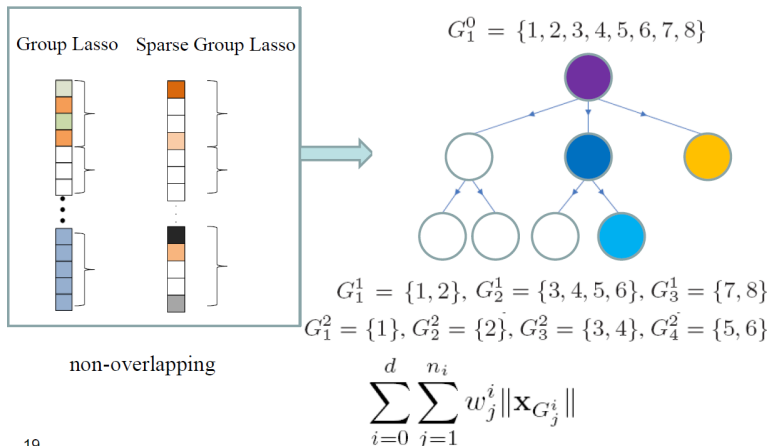
$$\sum_{g=1}^G \lambda_g \|\mathbf{x}_g\| + \lambda \|\mathbf{x}\|_1$$

② Fused LASSO:

$$\lambda_1 \sum_{k=1}^{K-1} \|x_k - x_{k+1}\| + \lambda_2 \|\mathbf{x}\|_1$$

③ Overlapped Group LASSO (groups share the same components)

Other Types of Sparse Modeling



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Figure 0.5: Illustration of Tree Based LASSO [J. Ye 10].